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Comparing Different *As-Treated* Approaches: A Methodological Study Using Real-World Data on Psoriasis Treatments

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ABSTRACT

Background: Accurate pharmaco-epidemiological assessment of the safety of biological treatments is essential but methodologically challenging.

Aims: This study evaluated how different assumptions regarding treatment switching and treatment episode definitions affect outcome estimates in psoriasis patients, focusing on malignancies and serious infections.

Methods: Data from Danish national registers (2009–2022) were analyzed for hospitalized psoriasis patients treated with ustekinumab, non-biologics, tumor necrosis factor-α inhibitors, and other interleukin inhibitors excluding ustekinumab. Various *as-treated* approaches were assessed and compared to the *Intention-to-Treat* (ITT) approach. The *as-treated* approaches included: (1) RC-switch, considering switching only between ustekinumab and the comparator in question, (2) Bio-switch, considering switches among all biologics, (3) Multi-switch, analyzing all groups simultaneously, and (4) Hierarchy-switch, analyzing switching by a predefined hierarchy. Hazard ratios (HRs) and 95% confidence intervals (CIs) were analyzed using Cox models. Sensitivity analyses incorporated treatment episodes based on the defined daily dose (DDD) and grace periods.

Results: Broader treatment switching allowances affected outcome estimates. For malignancies, the Bio-switch approach yielded the largest HR changes relative to the ITT. For serious infections, also, Bio-switch produced the largest HR shifts, for example, HR 1.32 (CI 1.09, 1.58) versus ITT HR 1.08 (CI 0.91, 1.29). Sensitivity analyses showed that accounting for episode gaps influenced HRs and widened CIs.

Conclusion: The number of events within the exposure group varied by approaches, influencing the HR estimates. The multi-switch approach effectively captured treatment switching and reduced statistical uncertainty, supporting clearer safety conclusions. Tracking each treatment period with gaps reflects real-world practice but may lower statistical power and should be carefully considered in analyses.

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Key Points

- The study highlights how various *as-treated* analytical approaches affect safety outcome estimates in psoriasis patients.
- Broader treatment switching considerations generally affected the results, for example the Multi-switch, analyzing all treatments simultaneously, most accurately reflected real-world treatment patterns and was especially effective in assessing serious infection risks.
- Sensitivity analyses showed that treatment episode definitions, including gaps and duration, significantly affect risk estimates, number of captured outcome events, and statistical precision.

Plain Language Summary

Understanding the safety of psoriasis medications is important but challenging because patients often switch between treatments. This study used data from Danish national health registers to examine how different ways of analyzing treatment switches affect estimated risks of cancer and serious infections. We compared several methods for handling treatment changes and found that the Multi-switch method, which considers all treatment changes together, most closely reflected real-life treatment patterns and provided clearer results, especially for infection risks. We also found that assumptions about treatment duration and gaps between treatments influence results and affect the reliability of the findings. This highlights the importance of careful choices in study design when evaluating the safety of medications.

1 | Introduction

The concept of estimands, as defined by the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) in its E9(R1) guideline, plays a pivotal role in epidemiological studies [1, 2]. Estimands delineate the specific aspects of treatment effects that researchers aim to measure and interpret. Among the various estimand approaches outlined in the guideline, the prominent methodology is the *Intention-to-Treat* (ITT) approach [1, 2].

The ITT approach has traditionally been considered the gold standard for Randomized controlled trials (RCTs) research for assessing efficacy and effectiveness [3, 4]. However, its applicability may be limited in scenarios characterized by multiple treatment options and instances of treatment switching, as it may attribute safety events to the initial treatment rather than the actual treatment being received [5–7]. Switching of psoriasis treatment is known to happen often, usually based on non-adequate treatment effects or unacceptable side effects; for example, approximately 14.4% of initial psoriasis treatments were switched to alternative treatments within 12 months due to severe side effects or other reasons, an aspect not captured by the ITT approach [8]. These complexities in real-world settings require alternative approaches that better capture the dynamic nature of treatment exposure and its effects [9–11].

In response to these challenges, the *as-treated* approach offers a more nuanced perspective, accounting for clinical realities such as treatment discontinuation and switching to alternative treatments [9, 12]. Unlike the *per-protocol* approach, which includes only those participants who strictly adhered to their originally assigned treatment and excludes those with major deviations, the *as-treated* analysis retains all participants and reclassifies them based on their actual exposure, thereby improving the safety evaluations [12–14]. The *as-treated* approach includes distinct strategies that track all treatments received during the study period and allow flexibility in treatment regimens, accommodating instances of treatment switching throughout the follow-up period. Some studies also adjust for treatment switching among multiple treatments to compare safety risks, particularly in psoriasis patients who may be exposed to various medications with treatment switching [9, 10].

Despite the advancements in the *as-treated* approach, certain critical aspects remain underexplored. Specifically, the influence of treatment switching patterns and the definition of treatment episodes on outcome risks warrant further investigation.

This study examines how different definitions of treatment switching patterns and gaps between treatment episodes affect the estimates in hospitalized psoriasis patients. Utilizing real-life data from the Danish national health registers encompassing all hospitalized patients diagnosed with psoriasis, we examined how variations in these assumptions influenced the risk of relevant outcomes. Our analysis, detailed in the method section below, aimed to enhance the understanding of the *as-treated* approach and its implications for real-world clinical practice.

2 | Methods

2.1 | Data Sources

This study was based on various data sources, including The National Patient Register [15], The National Prescription Register [16], The Danish Cancer Register [17], The Register of Causes of Death [18], and the Danish population registers [19]. Each Danish resident is assigned a unique Central Person Register (CPR) number [20], which enables linkage across these registers. The stability of the CPR number over time facilitates extensive longitudinal tracking of individuals across healthcare and administrative registers. The population registers provided information on demographics and migrations, while the National Patient Register offered detailed data on hospitalizations, outpatient visits, diagnoses, and procedures, including treatment with biological agents. Diagnoses were coded according to the International Classification of Diseases, version 10 (ICD-10) [21]. The Danish Cancer Register documented cancer diagnoses, the Cause of Death Register provided information on mortality, and The National Prescription Register contained records of dispensed prescriptions categorized by the WHO Anatomical Therapeutic Chemical classification system (ATC codes) [22].

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2.2 | Study Population

The study population comprised individuals with at least one hospital diagnosis of psoriasis or psoriatic arthritis (henceforth psoriasis) between 1994 and 2022, identified by specific ICD-10 codes: L40, M07.0, M07.1, M07.2, M07.3, and M09.0, including all subcodes [23]. Treatments administered before the first psoriasis diagnosis were excluded from the exposure definition.

2.3 | Exposure

Treatment exposure comprised psoriasis treatments used or introduced in Denmark from 2009 to 2022. Exposure groups were defined based on targeted molecules: interleukin (IL)-12/23 inhibitor (ustekinumab), non-biological systemic treatments (non-biological treatments), tumor necrosis factor-α inhibitors (TNF-α inhibitors), and other IL-targeted inhibitors, excluding ustekinumab (other-IL inhibitors). In the analyses, ustekinumab was used as the reference treatment group. Each treatment within these four categories was identified in the health registers by its ATC code or procedure code [20]. The complete list of ATC and procedure codes for the drugs in each treatment group is provided in the Supporting Information (Annex Table S1).

2.4 | Study Design

This nationwide, population-based cohort study spanned 14 calendar years (2009–2022) and employed an active-comparator new-user (ACNU) design [24]. New users were defined as individuals who initiated treatment with a drug from one of the exposure groups and had no record of the same drug substance in the preceding 365 days. The index date was defined as the date of dispensing or administration for patients meeting the inclusion criteria.

2.5 | Study Period

The exposure period began in 2009, coinciding with the introduction of ustekinumab in Denmark, which was the government-mandated "first-line" treatment for psoriasis [25]. At that time, TNF-α inhibitors and non-biological treatments were already available on the market. The exposure period for the other-IL inhibitors group commenced in 2015, following the approval of the first IL-17 inhibitor in Denmark. Individuals were followed until 31 December 2022. If a patient died or emigrated before the end of the study period, their individual follow-up time was censored. For each analysis, only the time to the first outcome event per individual was considered.

2.6 | Outcomes

This study assessed two clinically relevant outcomes in patients with psoriasis: malignancies excluding non-melanoma skin cancer (NMSC) and serious infections [26–28]. These outcomes were identified using ICD-10 codes, and a detailed list can be found in the Supporting Information (Annex Table S2).

Since malignancies typically develop slowly after potential exposure, a lag-time of 12 months was applied and all person-time at risk was shifted forward by 12 months relative to the recorded treatment episodes to account for the delayed development of malignancy outcomes (Annex Figure S1A) [29]. Thus, patients were excluded if a diagnosis of malignancy was recorded prior to exposure to their treatment or within 12 months following the first exposure, ensuring that only new cancer cases were included in the analyses (Annex Table S3).

The serious infections outcome represented more frequent diseases and was not subject to exclusions for prior events or any lag-time for the risk window (Annex Figure S1B).

2.7 | Various Approaches

This study investigated the impact of different assumptions regarding treatment switching during follow-up (Figure 1 and Table 1). For comparison, the ITT approach was applied. The ITT approach did not consider treatment switching and attributed subsequent events to the treatment group to which each individual was first exposed.

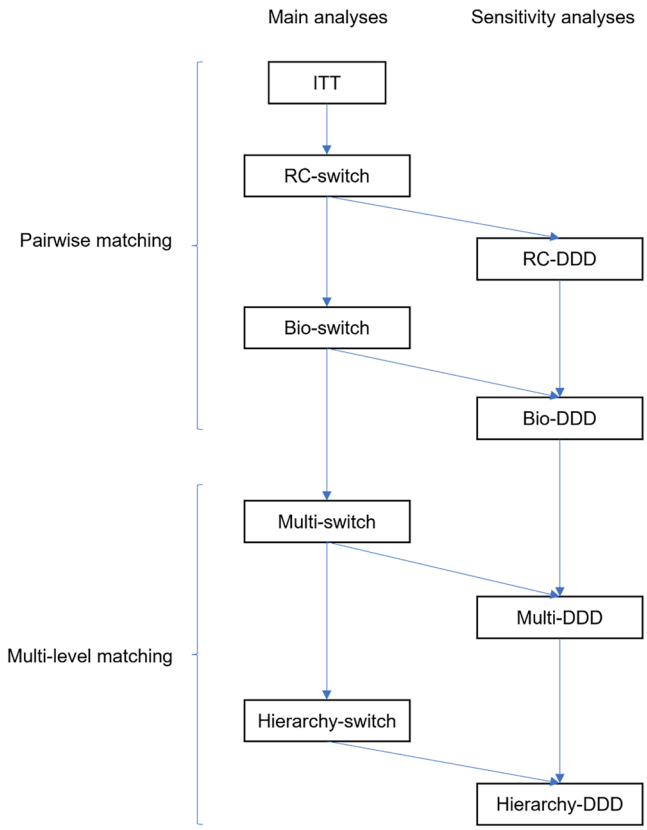

FIGURE 1 | Flowchart showing the steps of each approach. Modifications to each approach relative to the ITT approach are visualized. Vertical arrows represent changes in the definitions of treatment switching. Diagonal arrows represent sensitivity analyses in which a treatment episode was censored after the defined daily dose (DDD) plus a grace period (following the final treatment record in the series). Detailed information is provided in the Sensitivity analyses section. Each approach name corresponds to definitions provided in Table 1. The DDD-suffix denotes the sensitivity analysis of each approach.

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TABLE 1 | Overview of approaches.

Approach name	Treatment switching	Comparison groups
ITT	No	• Reference (ustekinumab) • One active comparator
RC-switch	Between reference and active comparator	• Reference (ustekinumab) • One active comparator
Bio-switch	Among biological treatments	• Reference (ustekinumab) • One active comparator • "Other or off-treatment"
Multi-switch	All treatments considered simultaneously	• Reference (ustekinumab) • TNF-α inhibitors • Other-IL inhibitors • Non-biological treatments
Hierarchy-switch	All treatments considered simultaneously, with hierarchical switching	• Reference (ustekinumab) • TNF-α inhibitors • Other-IL inhibitors • Non-biological treatments

Definitions of treatment switching were modified in steps (Figure 1). If a treatment switch occurred, the individual's exposure was reclassified as the most recent treatment. For Refence-comparator switch (RC-switch) and Bio-switch approaches, only one active comparator group was analyzed at a time. In contrast, the Multi-switch and Hierarchy-switch approaches included all three active comparator groups simultaneously (Table 1).

2.7.1 | RC-Switch

This approach accounted for treatment switching between the reference (ustekinumab) and the active comparator group in question. In contrast to the ITT approach, which ignored switches entirely, this approach reclassified exposure upon switching between these two groups.

2.7.2 | Bio-Switch

This approach accounted for all switches to any biological treatment, regardless of the comparator group in question. However, events occurring during these episodes were excluded from the relative comparison between ustekinumab and the comparator group in question (see "Other or off-treatment" in Table 1). If patients later switched back to ustekinumab or the active comparator group, their follow-up episodes could be re-included in the analysis.

2.7.3 | Multi-Switch

Similar to the Bio-switch approach, this approach accounted for all biologic treatment switches. However, all treatment episodes were included in the analysis since the exposure variable was generalized to include all four treatment groups.

Since non-biological treatments for psoriasis patients are often combined with biological treatments, switching to a non-biological treatment was only counted if the patient was the new user of the specific non-biological treatment.

2.7.4 | Hierarchy Switch

The Hierarchy-switch approach only considered treatment switching in accordance with a predefined hierarchy of treatment groups: Other-IL inhibitors > ustekinumab > TNF-α inhibitors > non-biological treatments. The hierarchy was based on drug approval dates.

To explain the implications of each approach, assume that the reference treatment group is ustekinumab and the active comparator is TNF-α inhibitors for a hypothetical Patient X (Figure 2). Under the ITT approach, Patient X is analyzed according to the initial treatment; therefore, Patient X contributed to the ustekinumab group regardless of any subsequent treatment changes. The RC-switch approach accounts for treatment switching only between the reference group and the active comparator, ignoring switches to other treatment groups (e.g., other-IL inhibitors). The Bio-switch approach considers all treatment switches, but if a switch to a treatment not included in the reference or comparator groups occurs, the patient is classified as "off-treatment". The Multi-switch approach treats all treatment switches equally, assigning the event according to the current treatment, such as the Other-IL inhibitors group. Finally, the Hierarchy-switch approach applies a predefined hierarchy of treatment groups to determine assignment when a switch occurs; in this case, the event is also assigned to the Other-IL inhibitors group.

2.8 | Matching

Ustekinumab users were matched with individuals from the active comparator group in question. Individuals were matched based on their propensity scores [30, 31]. The propensity scores summarized the probability of receiving a drug within the treatment group in question, based on individual characteristics such as sex, age, education, disease history, and recent treatment group (Annex Table S4). These scores were calculated using a logistic regression model specific to each calendar year of the study period. To create a matched dataset for statistical comparisons, relevant comparator patients were considered if their index date was within ±30 days of the index date of the new ustekinumab user.

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In addition, the study employed exact matching on the presence of a psoriatic arthritis diagnosis and the most recent treatment group before inclusion. A maximum of four comparator matches per ustekinumab user were included using a nearest neighbor approach. The ITT, RC-switch, and Bio-switch methods were based on this pairwise matching procedure (Figure 1). The framework was extended to a multi-level matching procedure for the Multi-switch and Hierarchy-switch approaches, which incorporated all four treatment groups within the same models.

The baseline characteristics of the matched patients are presented in Table 2. Detailed results of baseline characteristics are also provided in Annex Table S5. Covariate balance between treatment groups was assessed using absolute standardized mean differences (SMDs) before and after matching (Annex Figure S2). After matching, all covariates had absolute SMDs below 0.1.

2.9 | Sensitivity Analyses

In the main analyses, patients were allocated to the treatment group corresponding to their most recent exposure, following the different treatment switching rules described above. Additionally, each *as-treated* approach was subjected to sensitivity analyses (Figures 1 and 2). For the sensitivity analyses, a treatment episode was defined as starting from the treatment exposure date and extending through the defined daily dose (DDD, 12 weeks) plus a grace period of 30 days. The duration of the DDD and grace period were decided based on the standard treatment protocol of the reference drug, ustekinumab, in clinical practice [22]. Other active comparators were assigned the same DDD and grace period as ustekinumab to maintain consistency. If no subsequent treatment was registered by the end of the DDD plus the grace period, the treatment episode was deemed ended, and the individual was classified as being in an "off-treatment" state until initiation of a

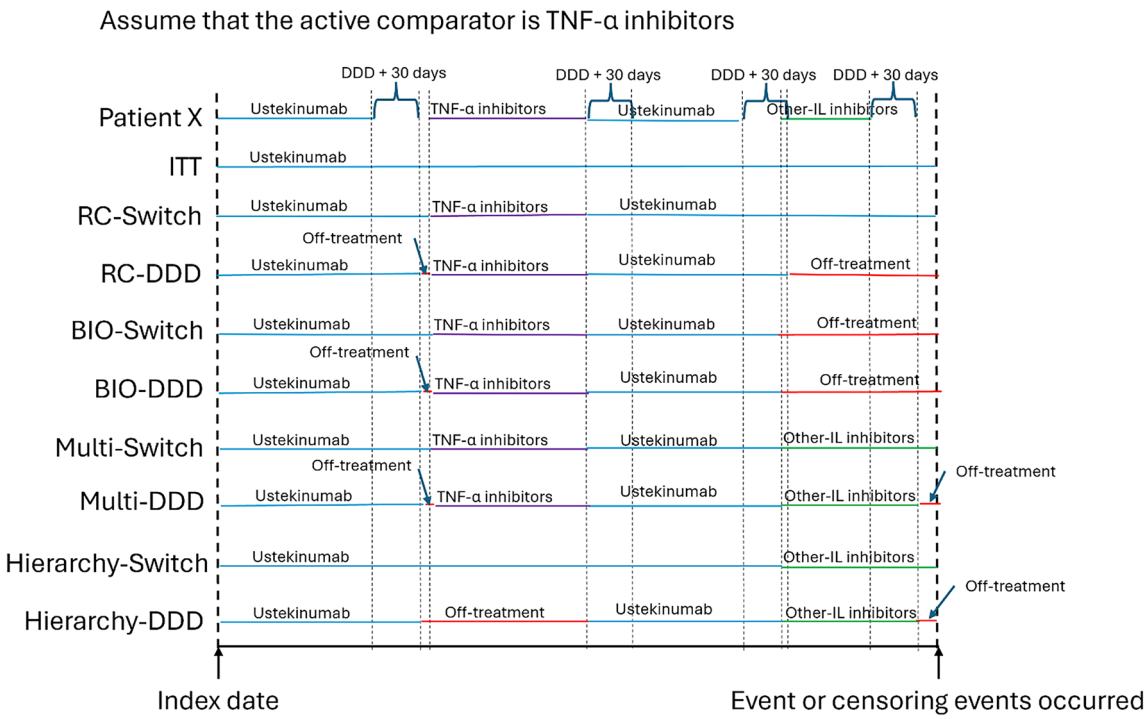

FIGURE 2 | Illustration of treatment episode construction by approach for a hypothetical patient. Each color represents treatment groups; Blue: ustekinumab, Purple: TNF-α inhibitors, Green: Other-IL inhibitors, and Red: Off-treatment. The names of the approaches refer to Table 1. The suffix "-DDD" indicates the sensitivity analyses of each approach.

TABLE 2 | Baseline characteristics of matched cohorts (*intention-to-treat* approach).

		Non-biological treatments		TNF-α inhibitors		Other-IL inhibitors	
Baseline characteristics		**ustekinumab**	**Non-biological treatments**	**ustekinumab**	**TNF-α inhibitors**	**ustekinumab**	**Other-IL inhibitors**
Total number of patients		1629	6475	1087	3830	493	1509
Sex	Male	923	3323	603	1933	249	787
	Female	706	3152	484	1897	244	722
Mean age (SD)		45.7 (15.43)	48.66 (17.17)	45.69 (15.19)	45.81 (15.47)	45.92 (17.27)	48.03 (14.4)

Abbreviation: SD, standard deviation.

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new episode. Events occurring during off-treatment were not presented in the analysis results. In the hypothetical case of Patient X (Figure 2), the sensitivity analyses (denoted with the suffix "DDD") follow the same switching rules as their corresponding main approaches, but treatment exposure is considered to end after the DDD plus a grace period of 30 days, after which patients are classified as "off-treatment".

2.10 | Statistical Analysis

For the *as-treated* analyses, Cox proportional hazards regression models with time-dependent exposure status were used to compare relative outcome rates between treatment groups. In the model, treatment switching under each approach was addressed by applying stabilized inverse probability of treatment weighting (IPTW) derived from propensity scores to minimize time-varying confounding after matching cohort, with ustekinumab defined as the reference treatment group (see "Matching" section and Annex Table S4) [32, 33]. Stabilized weights were then applied to balance treatment groups over time and account for differences in treatment exposure following switching. The hazard ratio (HR) measured the relative effect of each comparator group versus ustekinumab (the reference treatment), where HR <1 indicated a lower event rate in the comparator group, and HR >1 indicated a higher event rate. To present the uncertainty of the estimates, 95% CI were also reported.

2.11 | Statistical Programs and Packages

Statistical programming and data analyses were performed using R (version 4.3.1) [34] on a Windows 10 platform. Basic data management utilized the tidyverse (version 2.0.0) [35] and haven (version 2.5.3) [36] packages. Matching and statistical analyses were performed using the MatchIt (version 4.5.4) [37] and survival (version 3.5-5) [38] packages. Data visualization was created using ggplot2 (version 3.4.2) [39].

2.12 | Ethics Approval, Patient Consent Statement and Data Accessibility

The project was a register-based study. According to Danish law, ethical approval is not required for studies based solely on registry data [40]. Access to these data is restricted to authorized researchers through Statistics Denmark. Therefore, the present data collection is not publicly available, but similar data collections can be established by other researchers with appropriate authorization from Statistics Denmark.

3 | Results

3.1 | Results of Various Approaches

Increased flexibility in the definitions of treatment switching generally led to higher HRs across all comparator groups (Table 3, Figures 3 and 4). Notably, the Bio-switch approach was associated with the highest HRs for malignancies, while the Multi-switch approach yielded the highest HRs for serious infections.

For the malignancy outcome, the HRs for other-IL inhibitors showed the largest difference compared to the ITT approach and had the widest CIs. Specifically, the highest HR was 5.07 (CI 0.67, 38.15) under the Bio-switch approach, while the lowest HR was 1.37 (CI 0.72, 2.59) under the Hierarchy-switch approach, compared to 1.36 (CI 0.45, 4.07) for the ITT approach. Notably, the number of captured events varied by approach. For instance, under the ITT approach, 4 events were identified among 402 at-risk individuals treated with ustekinumab, whereas the Bio-switch approach captured only 1 event among 627 at-risk individuals.

For the more frequent outcome of serious infections, the CIs from the *as-treated* approaches were wider compared to those based on the ITT approach. However, the changes in both HRs and their CIs were smaller than those observed for the malignancy outcome. Specifically, the HR for TNF-α inhibitors under the ITT approach was 1.08 (CI 0.91, 1.29), while the highest HR was 1.32 (CI 1.09, 1.58) under the Bio-switch approach. Notably, the interpretation shifted from a non-significant to a statistically significant result depending on the analytical approach used.

3.2 | Sensitivity Analyses

Applying the DDD plus a 30-day grace period to construct treatment episodes in the sensitivity analyses resulted in higher HRs with wider CIs compared to the original analyses (Table 4, Figures 3 and 4). For example, for the malignancy outcome, the HR for non-biological treatments was 1.84 (CI 1.07, 3.15) under the RC-DDD approach compared to 0.99 (CI 0.68, 1.44) under the original RC-switch approach.

The sensitivity analysis results for serious infections showed a similar trend to that observed for malignancies (Table 4). Notably, the HR for the TNF-α inhibitors was 1.36 (CI 1.09, 1.69) under the Hierarchy-DDD approach, compared to 1.17 (CI 0.97, 1.4) under the Hierarchy-switch approach.

4 | Discussion

In this comprehensive national cohort study, we investigated the effects of different *as-treated* approaches, each applying varying definitions of treatment switching, over a 14-year follow-up period (2009–2022) among Danish psoriasis patients. We found that the choice of approach directly influenced the number of outcome events captured following treatment exposures, which in turn notably affected the estimated HRs and CIs. The results were also sensitive to the application of the DDD plus a grace period for defining treatment episodes, as this approach did not capture events occurring during gaps between episodes. These methodological choices had a more pronounced impact on the malignancy outcome than on serious infections, likely due to the lower incidence and longer latency period of malignancies.

4.1 | Various Approaches

Variations in results across different approaches largely depended on the number of outcome events captured during the exposure period of the current treatment. In real-world settings,

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TABLE 3 | Hazard ratios for malignancies and serious infections comparing ustekinumab and each of the three comparator groups.

		Comparators[b]					
		Non-biological treatments		TNF-α inhibitors		Other-IL inhibitors	
Outcome	**Approach[a]**	**HR (CI)**	**Events/patients (ustekinumab)**	**HR (CI)**	**Events/patients (ustekinumab)**	**HR (CI)**	**Events/patients (ustekinumab)**
Malignancies	ITT	0.74 (0.51, 1.05)	114/5773 (40/1486)	0.75 (0.49, 1.15)	76/3387 (29/971)	1.36 (0.45, 4.07)	16/1221 (4/402)
	RC-switch	0.99 (0.68, 1.44)	117/6553 (36/1955)	1.15 (0.73, 1.81)	77/4234 (25/1576)	3.03 (0.7, 13.04)	18/1558 (2/628)
	Bio-switch	1.44 (0.88, 2.36)	103/6543 (19/1920)	1.46 (0.87, 2.45)	75/4230 (18/1564)	5.07 (0.67, 38.15)	16/1549 (1/627)
	Multi-switch	1.3 (0.78, 2.17)	69/4793 (19/1901)	1.04 (0.58, 1.87)	27/3062 (19/1901)	1.66 (0.82, 3.38)	13/1123 (19/1901)
	Hierarchy-switch	1.17 (0.74, 1.85)	65/4304 (26/1872)	0.82 (0.46, 1.44)	22/2452 (26/1872)	1.37 (0.72, 2.59)	15/1131 (26/1872)
Serious infections	ITT	0.89 (0.77, 1.02)	872/6475 (248/1629)	1.08 (0.91, 1.29)	604/3830 (163/1087)	0.87 (0.64, 1.18)	147/1509 (57/493)
	RC-Switch	1.01 (0.87, 1.16)	868/7446 (246/2161)	1.24 (1.05, 1.48)	590/4770 (164/1690)	0.82 (0.6, 1.1)	146/1718 (58/681)
	Bio-switch	1.09 (0.92, 1.29)	724/7331 (166/2084)	1.32 (1.09, 1.58)	558/4742 (140/1664)	1.04 (0.73, 1.49)	129/1698 (39/669)
	Multi-switch	1.1 (0.92, 1.32)	533/5348 (159/2044)	1.29 (1.06, 1.57)	270/3266 (159/2044)	1.22 (0.92, 1.61)	70/1107 (159/2044)
	Hierarchy-switch	0.95 (0.81, 1.12)	485/4838 (214/2078)	1.17 (0.97, 1.4)	248/2665 (214/2078)	1.1 (0.86, 1.42)	85/1127 (214/2078)

Note: Events/Patients, the number of events divided by the number of individuals at risk.
Abbreviations: 95% CI, confidence interval; HR, hazard ratio.
[a]The name of approaches refers to Table 1.
[b]Non-biological treatments and TNF-α inhibitors were followed from 2009 to 2022, and the other IL inhibitor group was followed from 2015 to 2022.

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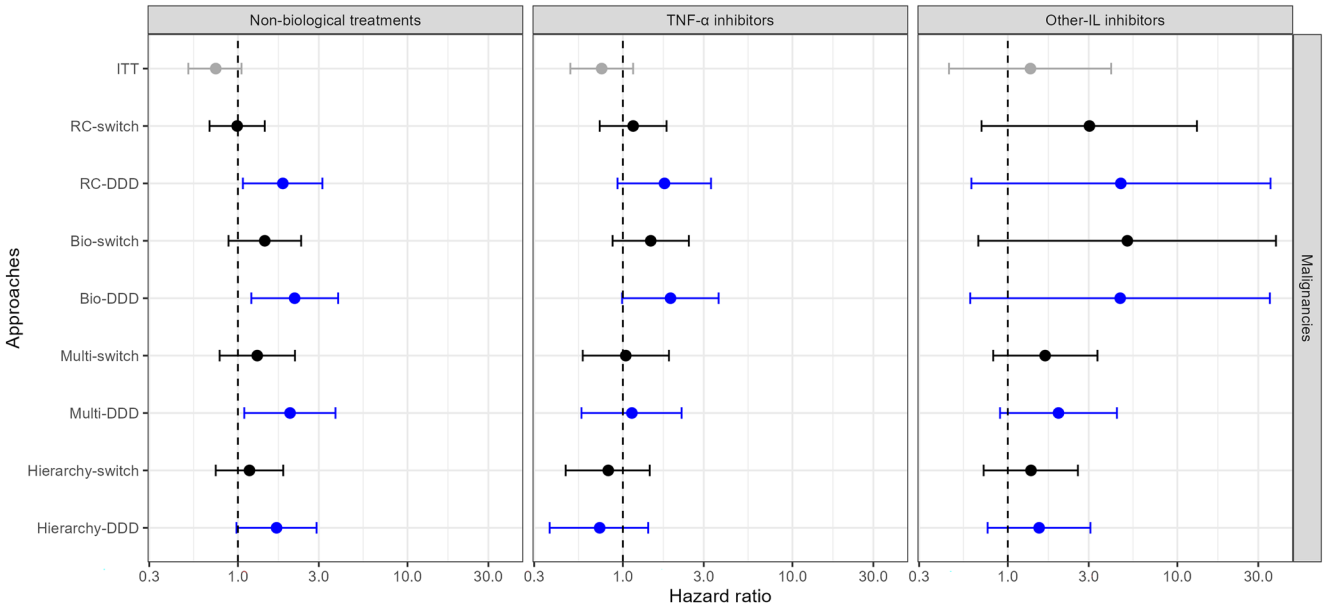

FIGURE 3 | Hazard ratios for malignancies by various approaches and comparator groups. Dots represent the point estimates of hazard ratios (HR) comparing ustekinumab to each comparator. Horizontal lines show the 95% confidence intervals. The black vertical dashed lines indicate equal hazards between ustekinumab and the comparator. Colors denote the approaches: gray for ITT, black for *as-treated* approaches, and blue for sensitivity analyses. HR < 1 favors the comparator, while HR > 1 favors ustekinumab.

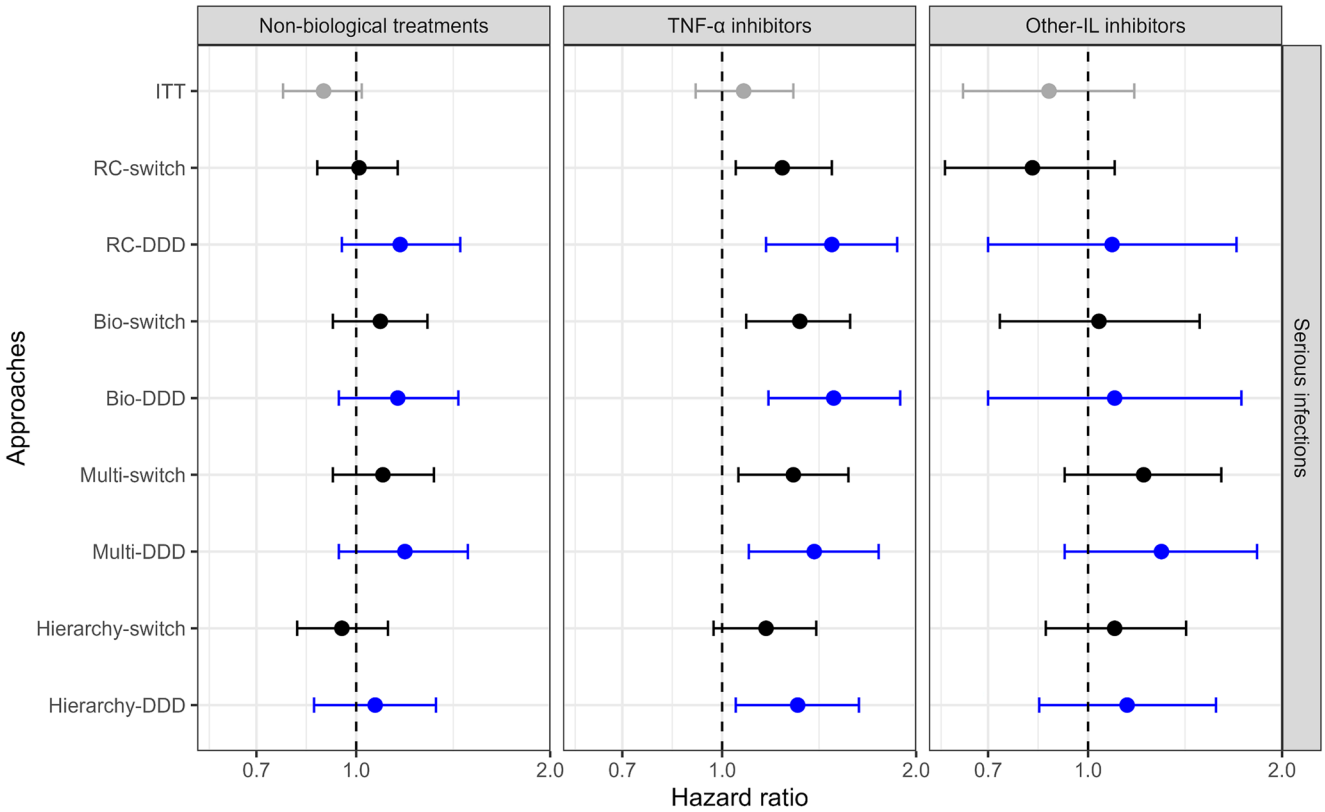

FIGURE 4 | Hazard ratios for serious infections by various approaches and comparator groups. Dots represent the point estimates of hazard ratios (HRs) comparing ustekinumab to each comparator group. Horizontal lines show the 95% confidence intervals. The black vertical dashed lines indicate equal hazards between ustekinumab and the comparator. Colors denote the approaches: gray for ITT, black for *as-treated* approaches, and blue for sensitivity analyses. Blue color indicates sensitivity analyses. HR < 1 favors the comparator, while HR > 1 favors ustekinumab.

treatments for complex diseases like psoriasis often change due to various factors such as national guidelines, adverse effects, or lack of treatment effect.

Under the ITT principle, the initial treatment is assumed to continue until the end of follow-up, regardless of any subsequent treatment changes. Thus, the ITT approach may wrongly

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TABLE 4 | Hazard ratios from sensitivity analyses for malignancies and serious infections comparing ustekinumab and each of the three comparator groups.

		Comparators[b]					
		Non-biological treatments		TNF-α inhibitors		Other-IL inhibitors	
Outcome	**Approach[a]**	**HR (CI)**	**Events/patients (ustekinumab)**	**HR (CI)**	**Events/patients (ustekinumab)**	**HR (CI)**	**Events/patients (ustekinumab)**
Malignancies	RC-DDD	1.84 (1.07, 3.15)	62/6550 (17/1939)	1.76 (0.93, 3.31)	51/4211 (12/1565)	4.64 (0.61, 35.43)	13/1554 (1/628)
	Bio-DDD	2.16 (1.2, 3.9)	56/6519 (14/1917)	1.91 (0.99, 3.67)	51/4209 (11/1559)	4.6 (0.6, 35.11)	13/1547 (1/627)
	Multi-DDD	2.03 (1.09, 3.76)	38/4615 (14/1898)	1.13 (0.57, 2.22)	21/3056 (14/1898)	1.99 (0.9, 4.4)	11/1121 (14/1898)
	Hierarchy-DDD	1.69 (0.98, 2.91)	35/4283 (21/1869)	0.73 (0.37, 1.41)	15/2445 (21/1869)	1.53 (0.76, 3.07)	13/1130 (21/1869)
Serious infections	RC-DDD	1.17 (0.95, 1.44)	365/7117 (124/2080)	1.52 (1.2, 1.92)	371/4610 (90/1648)	1.09 (0.7, 1.71)	82/1671 (26/671)
	Bio-DDD	1.15 (0.93, 1.43)	337/7097 (119/2076)	1.53 (1.21, 1.94)	368/4607 (88/1646)	1.11 (0.7, 1.74)	80/1671 (25/670)
	Multi-DDD	1.18 (0.94, 1.48)	241/4947 (113/2040)	1.45 (1.15, 1.83)	206/3249 (113/2040)	1.3 (0.92, 1.83)	47/1097 (113/2040)
	Hierarchy-DDD	1.07 (0.86, 1.33)	223/4628 (144/2026)	1.36 (1.09, 1.69)	186/2616 (144/2026)	1.15 (0.84, 1.58)	54/1104 (144/2026)

Note: Events/Patients: the number of events divided by the number of individuals at risk.
Abbreviations: 95% CI, confidence interval; HR, hazard ratio.
[a]The name of approaches refers to Table 1. The suffix "-DDD" indicates the sensitivity analyses of each approach.
[b]Non-biological treatments and TNF-α inhibitors were followed from 2009 to 2022, and other IL inhibitors were followed from 2015 to 2022.

attribute events occurring during the follow-up to the index treatment of the patient.

Different rules for defining treatment switching during follow-up influenced the estimated HRs. In the RC-switch approach, which accounted for switching between ustekinumab and the active comparator in question, only a small number of events were attributed to another treatment group, resulting in HRs similar to those of the ITT approach. In contrast, the Bio-switch approach accounted for switching among the other biological treatments, which led to the exclusion of events that occurred after switching, reducing the number of events included in the comparisons.

The Multi-switch approach included all treatment groups in the same analysis, considering events to be attributed to the current treatment group, thereby retaining more events in the analysis. The hierarchy-switch approach, which used the approval dates of the study drugs to guide treatment switching, was more conservative, yielding results more similar to the ITT approach, as many treatment switches were disregarded.

4.2 | Sensitivity Analyses

The results were clearly sensitive to assumptions regarding treatment episode duration based on DDD plus grace period. Applying these assumptions led to wider CIs and, in some cases, even reversed the direction of the HRs. The increased uncertainty was due to fewer events being captured, as gaps between treatment episodes led to the exclusion of a number of events from the analysis.

4.3 | Limitations

Confounding is an inherent challenge in observational studies. We addressed this through propensity score matching. Also, time-varying confounding was addressed by IPTW. However, the possibility of unmeasured confounding cannot be ruled out completely. Another limitation of the study was the variation in registration practices among Danish dermatology departments, which may have influenced results in either direction. Even if such misclassification was non-differential, it could still have biased the findings toward no difference between treatment groups.

Although we applied a lag-time for malignancy outcomes to account for their typically long latency period, malignancies can be due to various factors, meaning the setting may not fully reflect real-world dynamics. The chosen lag-time could also influence the number of captured events. Additionally, patients may experience treatment gaps longer than the defined grace period for reasons such as missed appointments, which could affect the number of events attributed to off-treatment.

A known limitation is the reliance on Cox models due to possible lack of proportional hazards and depletion of susceptible individuals [41, 42]. Thus, our results should be interpreted as measures of association for the relation between the treatment groups and the safety outcomes.

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5 | Conclusion

Accurate pharmaco-epidemiological estimation of biological treatment safety is essential for patients and healthcare providers. However, statistical modeling, particularly in defining relevant estimands, can be challenging. In clinical practice, treatment switching is common, and the ITT approach is not optimal for evaluating the safety of treatments in the presence of treatment switching. Our study findings suggest that different *as-treated* approaches offer distinct advantages, with the Multi-switch approach, which analyzes all treatment groups at the same time with the same weight, providing the most realistic representation of clinical practice. Assumptions regarding treatment episode durations, such as DDD and a grace period, should reflect real-world clinical settings while addressing gaps between treatment episodes. In summary, the Multi-switch approach, combined with appropriate DDD and grace period considerations, best captures treatment safety in real-world settings.

Author Contributions

Dr. Kim contributed to the research design, data management, statistical analyses, data visualization, and the writing and editing of the manuscript. Dr. Jensen was responsible for conceptualizing the study, refining the research design, overseeing data collection, conducting statistical analyses, and editing the manuscript. Prof. Egeberg handled result validation, integrated clinical insights into the document, and contributed to manuscript editing. Prof. Stensballe provided study management and direction, helped conceptualize the study, and edited the manuscript. All authors have read and agreed to the published version of the manuscript.

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The authors have nothing to report.

Conflicts of Interest

Prof. Egeberg has received research funding from Almirall, Pfizer, Eli Lilly, Novartis, Bristol-Myers Squibb, AbbVie, Janssen Pharmaceuticals, Boehringer Ingelheim, the Danish National Psoriasis Foundation, the Simon Spies Foundation, and the Kgl Hofbundtmager Aage Bang Foundation, and honoraria as consultant and/or speaker from Amgen, AbbVie, Almirall, Leo Pharma, Zuellig Pharma Ltd., Galápagos NV, Sun Pharmaceuticals, Samsung Bioepis Co. Ltd., Pfizer, Eli Lilly and Company, Novartis, Union Therapeutics, Galderma, Dermavant, UCB, Mylan, Bristol-Myers Squibb, McNeil Consumer Healthcare, Horizon Therapeutics, Boehringer Ingelheim, and Janssen Pharmaceuticals, and is currently employed by LEO Pharma. Prof. Stensballe has received research funding from AbbVie. Dr. Kim and Dr. Jensen have no conflicts of interest.

Data Availability Statement

Access to the data is restricted to authorized researchers through Statistics Denmark. Therefore, the present data collection is not available to others, but similar data collections can be established by other researchers with Statistics Denmark authorization.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Illustration of lag time applied to the exposure episodes for disease outcomes. Solid lines along the time-at-risk axis represent person-time included in the analysis, whereas dashed lines indicate censored exposure periods. (A) Application of lag time to the exposure period for malignancies. (B) Definition of the exposure period for serious infections. **Figure S2:** Covariate balance assessment through matching. The blue dots represent absolute standardized mean difference (SMD) of each covariate before the matching. The red dots represent absolute SMD of each covariate after matching. The black vertical line indicates the point of 0 of absolute SMD of each covariate. **Table S1:** ATC codes and procedure codes of each drug used in psoriasis treatment. **Table S2:** ICD-10 codes used to define the outcomes. All subcodes were included. **Table S3:** The number of patients for each treatment by exclusion criteria* of the malignancy outcome. **Table S4:** Baseline characteristics for propensity scores. All subcodes are included. **Table S5:** Detailed baseline characteristics of matched cohorts (intention-to-treat approach).

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